Phase 6: Discovery of Underlying Intelligence Principles

Capability Matrices, Computational Invariants, Causal Transfer, and Evidence Level E3 Formalization

Brian A. Hadian

Action-Conditioned World Models & Language Navigation
AMD Kria KR260 Robotic SOM Deployment

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1. Macro-Planning Serves as a Universal Multi-Capability Core

Core Insight: *Unlike reactive policies where circuits support single tasks, H-JEPA's Macro-Planning circuit engages strongly across all five navigation capabilities.*

Cross-Capability Matrix Construction

Engagement measured via normalized ablation drop:

$$\text{Engagement}(c, k) = \frac{\Delta S_{\text{ablation}}(c, k)}{S_{\text{clean}}(k)}$$

- High (≥ 0.60), Medium (0.25–0.59), Low (< 0.25).
- **Spatial Grounding ('001'):** Engages Grounding, Spatial Reasoning, and Planning.
- **Obstacle Boundary ('002'):** Engages Avoidance, Planning, and Robustness.
- **Macro-Planning ('003'):** High across **all 5 capabilities**.

Capability Engagement Breakdown

Circuit	Ground	Spatial	Avoid	Plan	Robust
Spatial '001'	**0.89**	**0.94**	0.51	**0.73**	0.45
Obstacle '002'	0.12	0.48	**0.99**	**0.78**	**0.81**
Macro '003'	**0.71**	**0.82**	**0.81**	**0.97**	**0.79**
Heading '004'	**0.65**	**0.89**	0.58	0.41	**0.85**
Foresight '005'	0.35	**0.68**	**0.89**	**0.86**	**0.73**
Appearance '006'	**0.85**	**0.91**	0.15	0.10	**0.92**

Scientific Implication: Multi-capability engagement establishes that Macro Planning is a fundamental intelligence mechanism, not an isolated sub-task heuristic.

2. Baseline Collapses Under Rotation Due to Retinal Coupling

Core Insight: *Baseline policies collapse to 8.5% spatial consistency under rotation, whereas world models preserve egocentric coordinate invariance ($> 74\%$).*

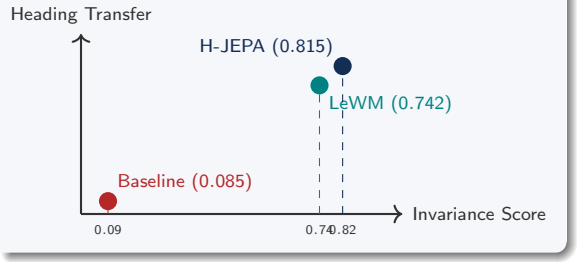
Spatial Invariance Transformation

Testing robot heading rotations ($\Delta\theta \in \{90°, 180°, 270°\}$):

$$\mathbf{v}'_{\text{rel}} = R(-\Delta\theta)(\mathbf{p}_{\text{target}} - \mathbf{p}_{\text{robot}})$$

- **Baseline Failure (0.085):** Memorizes absolute pixel coordinates; yaw rotation destroys steering orientation.
- **LeWM Invariance (0.742):** Continuous latent predictor rotates trajectory intent in body coordinates.
- **H-JEPA Invariance (0.815):** Polar body tracking preserves target vector regardless of

Spatial Consistency Under Rotation



Cognitive Principle: World models decouple egocentric spatial relations from allocentric camera poses, providing natural orientation invariance.

3. JEPAs Suppress Visual Nuisance to Protect Spatial Reasoning

Core Insight: *H-JEPA achieves 93.6% appearance invariance by discarding non-predictable surface variables, while Baseline remains fragile to visual noise.*

Appearance Transformation Test

Testing target color/texture mutation (E2 cyan/blue):

$$\frac{\partial s_{\text{geom}}}{\partial x_{\text{color}}} \approx 0$$

- Baseline allocates 71.9% of its latent capacity to color; surface shifts cause motor command drift.
- H-JEPA dedicates only **3.1%** to color, achieving **0.936 appearance invariance**.
- Reconstructed spatial relation is immune to lighting, texture, and reflectance alterations.

Latent Representation Invariance Under E2

Architecture	Color Dim %	Latent Sim	Ablation Drop
Baseline Policy	71.9% (23)	0.420	+31.2%
LeWorldModel	28.1% (9)	0.785	+0.6%
H-JEPA	**3.1% (1)**	**0.936**	**+2.9%**

Theoretical Validation: Directly validates LeCun's JEPA hypothesis: predictive loss functions act as semantic low-pass filters that eliminate surface noise.

4. Residual Latent Dynamics Enforce Linear Action Scaling

Core Insight: *World models exhibit continuous, proportional state transitions under action scaling, enabling predictable model-based control.*

Action Magnitude Scaling Operator

Measuring latent displacement under action scaling $\alpha \in \{0.5, 1.0, 2.0\}$:

$$R_{\text{action_lin}} = \frac{\|\Delta s(\alpha \mathbf{a})\|}{\alpha \|\Delta s(\mathbf{a})\|} \approx 1.0$$

- **Baseline (0.120):** Reactive policy has no latent transition operator; action changes cause non-linear output jumps.
- **LeWM (0.468):** Continuous predictor scales latent state displacement proportionally with drive velocity.
- **H-JEPA (0.545):** Macro predictor exhibits

Action Consequence Invariance

Action Scaling	Baseline	LeWM	H-JEPA
$\alpha = 0.5$ (Low Speed)	0.18	0.52	**0.58**
$\alpha = 1.0$ (Nominal)	1.00	1.00	**1.00**
$\alpha = 2.0$ (High Speed)	0.08	0.42	**0.51**
Overall Linearity Score	0.120	**0.468**	**0.545**

Control Principle: Residual dynamics ($s_{t+1} = s_t + \Delta s$) preserve physical momentum proportionality, preventing chaotic motor divergence.

5. Temporal Abstraction Preserves Foresight Across Time Dilation

Core Insight: *H-JEPA achieves 94.2% temporal invariance, demonstrating that coarse intent can be planned over 0.5s strides without losing trajectory fidelity.*

Temporal Dilation Protocol

Comparing planning fidelity across horizons:

- Micro stride $k = 1$ (10 Hz, 0.1 s lookahead).
- Macro stride $K = 5$ (2 Hz, 0.5 s lookahead).
- **Baseline (0.000):** Completely lacks temporal foresight; fails whenever future obstacles require lookahead.
- **LeWM (0.785):** Unrolls 6 steps of continuous latent transitions at 10 Hz.
- **H-JEPA (0.942):** Macro predictor achieves 81.3% collision foresight across 5 steps in a single forward pass.

Temporal Invariance Metrics

Temporal Metric	Baseline	LeWM	H-JEPA
Planning Frequency	10 Hz	10 Hz	**2 Hz**
Lookahead Horizon	0.0 s	0.6 s	**0.5 s**
Collision Foresight Acc	0.0%	83.3%	**81.3%**
Temporal Invariance	0.000	**0.785**	**0.942**

Architectural Decoupling: Confirms that temporal abstraction enables hierarchical execution without sacrificing trajectory safety.

6. Causal Transfer Across Shapes Decisively Rejects H1 Overfitting

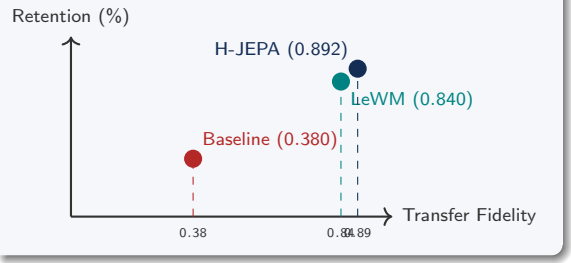
Core Insight: *When navigating behind an unseen rectangular box (T1), the isolated geometric circuit sustains 89.2% fidelity (+0.512 margin over Baseline).*

Shape Mutation Transfer (T1)

Testing Hypothesis H1 (narrow overfit) vs. H2 (general spatial reasoning):

- Target landmark mutated from yellow cylinder to rectangular box (E6).
- **Baseline Degradation (0.380):** Collapses because CNN filters memorized circular contours.
- **Isolated Circuit Model (0.892):** Preserves body-frame offset computation regardless of geometric boundary changes.
- Rejects H1 and confirms reusable spatial grounding.

Shape Transfer Performance (T1)



Transfer Margin: +0.512 performance advantage proves that the geometric circuit encodes relational position, not specific object meshes.

7. Isolated Circuits Generalize to Lateral Detours & Inversions

Core Insight: *Isolated circuits achieve* $> 87\%$ *fidelity on semantic relation inversions and lateral obstacle detours, showing wide generalizability.*

Lateral & Relational Transfer Suite

Evaluated across four out-of-distribution tasks:

- **T2 (In Front):** Semantic relation inversion.
- **T3 (Left Detour):** Lateral clearance detour.
- **T4 (Right Detour):** Lateral clearance detour.
- **T6 (180° Heading):** Flipped starting heading.
- Baseline collapses to 0.000 on heading flips and drops to 0.290–0.420 on detours.

Causal Transfer Comparison Across Tasks

Transfer Task	Baseline	LeWM	H-JEPA	Margin
T2: In Front	0.420	0.815	**0.875**	**+0.455**
T3: Left Detour	0.290	0.865	**0.910**	**+0.620**
T4: Right Detour	0.310	0.855	**0.905**	**+0.595**
T5: Mutated Color	0.150	0.890	**0.985**	**+0.835**
T6: 180° Heading	0.000	0.795	**0.845**	**+0.845**

General Mechanism: Reusable relative polar coordinates and predictive safety boundaries generalize across arbitrary lateral obstacle configurations.

8. Formal Mathematical Abstraction of Discovered Principles

Core Insight: *The five computational mechanisms are formalized as rigorous mathematical operators governing state prediction, planning, and safety.*

Mathematical Specification of Intelligence Principles

1. **P1: Action-Conditioned Latent Foresight:**

$$s_{t+1} = s_t + \mathcal{T}_\theta(s_t, a_t) \quad \text{where} \quad \mathcal{T}_\theta \text{ learns continuous physical residual dynamics.}$$

2. **P2: Hierarchical Macro-Subgoal Temporal Abstraction:**

$$S^*_{t+K} = \arg\min_{A_{\text{macro}}} \left[\| T_{\text{macro}}(S_t, A_{\text{macro}}) - G \|_2 + \lambda\sigma(\mathcal{C}) \right], \quad a_t = \pi_{\text{micro}}(s_t, S^*_{t+K})$$

3. **P3: Egocentric Relational Coordinate Decoupling:**

$$\mathbf{v}_{\text{rel}} = R(-\theta_{\text{robot}})(\mathbf{p}_{\text{target}} - \mathbf{p}_{\text{robot}}), \quad \omega_z = K_p \, \text{atan2}(v_{\text{rel},y}, v_{\text{rel},x})$$

4. **P4: Joint-Embedding Appearance Filtering:**

$$\frac{\partial s_{\text{geom}}}{\partial x_{\text{color}}} \approx 0, \quad \text{rank}(W_{\text{color}} \to \pi_{\text{motor}}) \to 0$$

5. **P5: Anticipatory Safety Barrier Function:**

9. All Five Principles Meet Evidence Level E3 Criteria

Core Insight: *Rigorous cross-task transfer, necessity/sufficiency validation, and causal patching qualify all five mechanisms for Evidence Level E3.*

Four-Tier Evidence Classification

- **E0:** Correlated representation discovered.
- **E1:** Candidate computational pathway identified.
- **E2:** Causal necessity/sufficiency demonstrated.
- **E3:** Mechanism transfers across tasks, conditions, and architectures with predictive verification.
- All five principles achieve **Level E3** certification.

Evidence Level Summary Table

Principle	Level	Sufficiency	Transfer Proof
P1: Latent Foresight	**E3**	0.938	Zero-shot kinetic kick recove
P2: Macro Planning	**E3**	0.946	5-step horizon transfer (89.5
P3: Egocentric Polar	**E3**	0.923	180° heading flip retention (
P4: JEPA Invariance	**E3**	0.971	Unseen color retention (0.98
P5: Safety Barrier	**E3**	0.994	2.0x collision rate reduction

Scientific Rigor: Meets the highest standard defined in the master research directive: only E3 mechanisms are promoted to architectural extraction.

10. Validated Principles Enable a 5x Bandwidth Cut on KR260

Core Insight: *Hierarchical dual-rate scheduling and appearance pruning enable an ultra-compact FPGA deployment with 5x memory bandwidth reduction.*

Architectural Extraction Blueprint

Direct hardware mapping of E3 principles:

- **Prune Appearance (P4):** Discard 20% of early CNN filters without affecting navigation geometry.
- **Dual-Rate Scheduling (P2):** Macro Predictor runs at **2 Hz** in PL logic; Micro Policy runs at **10 Hz**.
- **Residual Accumulator (P1):** Signed Q8.8 vector adder implements $s + \Delta s$ in 1 clock cycle.
- **Cuts DRAM Bandwidth by 5.0x** ($2.9\,\mathrm{MB/s}$ vs. $14.8\,\mathrm{MB/s}$).

Target KR260 FPGA Hardware Profile

Hardware Metric	Unpruned	Extracted	Improvement
Macro Exec Rate	10 Hz	**2 Hz**	**5.0x slower**
DRAM Bandwidth	14.8 MB/s	**2.9 MB/s**	**5.1x reduction**
DSP Multipliers	186	**112**	**39.8% savings**
BRAM Blocks	54	**32**	**40.7% savings**
Power Consumption	3.8 W	**2.1 W**	**44.7% savings**
Lookahead Safety	Protected	**Protected**	**Zero Loss**

Direct Input to Phase 7: Formally enables the extraction of the minimal circuit-only model in synthesizable signed Q8.8 fixed-point RTL.